

91. Heterostasis is characterized by:
- Hypervascularization
 - Hyperventilation
 - Tachycardia
 - ☒ All of the above
 - None of the above
92. Adaptation to cold environment is characterized by all of the following except:
- Reduced basal metabolic rate
 - Increased thermal conductivity
 - A skin temperature of 12-15 C
 - Increased subcutaneous fat
 - Decrease in surface area of the body
93. Which of the following is an example of behavioral response:
- Regulation of arterial blood pressure
 - Regulation of body temperature
 - Regulation of fluid volume
 - Control of diet
 - Release of catecholamines
94. A physiological error signal is:
- The same as a reference value
 - Transmitted to a detector via efferent nerves
 - Sent to effector organs via afferent nerve
 - ☒ The difference between a reference value (set point) and a new value signaled to an effector organ
 - None of the above
95. The rennin-angiotensin-aldosterone system (RAAS) is required in the regulation of:
- Arterial blood pressure
 - Blood volume
 - Fluid osmolality
 - H⁺ and K⁺ secretion
 - All of the above
96. In the establishment of the mechanisms of physiological adjustment Hans Selye:
- Explained the scientific study of stress
 - Coined the term "homeostasis"
 - Stated that homeostasis favours stability
 - Described "illhealth vs. imbalance"
 - Described the relationship between the Autonomic nervous system and fight flight reactions.
97. Multiplicity of receptors and effectors:
- Ensures stability of regulation and survival
 - Occurs in the cardiovascular system
 - Contributes to continuity
 - All of the above are true
 - Leads to instability of internal environment

98. Allostasis

- a. Is attaining viability through acute change
- b. Is adjustment to new conditions through chronic change
- c. When overloaded, may result in excessive stress hormone exposure
- d. May end up with atherosclerosis when elevation of arterial BP is persistently elevated
- e. May be characterized by all of the above except "b".

Select A. if 1,2,3 are correct
 B. if 1,3 are correct
 C if 2 and 4 are correct
 D if only 4 is correct
 E if all four are correct

99. Electrolyte excess/is characterized by:

- 1. Headache and vomiting
- 2. Watery diarrhea
- 3. Excessive drinking
- 4. Thirst

100. Behavioral responses may be exemplified by:

- 1. Regulation of blood glucose
- 2. Regulation of arterial blood pressure
- 3. Regulation of body temperature
- 4. Control of environmental condition

101. Match:

- | | |
|--------------------------------|-------------------------------------|
| 1. Control of diabetes by diet | a) Indicator of allostatic overload |
| 2. Control of body temperature | b) Behavioural response |
| 3 Positive feedback | c) Replication of micro-organisms |
| 4. Negative feedback | d) Reflex mechanism |
| 5. Heterostasis | e) A feedback factor of 5 |
| 6. Metabolic deterioration | f.) Hypervascularization |

T/F

102. In the ontogeny of regulatory mechanisms:*

- a. Survival precedes
- b. Regulation in time and amount is the last process
- c. Linked functional processes come second
- d. Structural specialization comes first

103. The regulation of the level of glucose in the blood involves

- a. The hypothalamus as a coordinating centre
- b. The hypothalamus as an effector organ
- c. The Liver as a receptor organ
- d. Endocrine pancreas as a receptor organ

ADDIS ABABA UNIVERSITY
COLLEGE OF HEALTH SCIENCES
SCHOOL OF MEDICINE

MODULE 1 FINAL EXAMINATION FOR FIRST YEAR MEDICAL STUDENTS

Friday, May 16, 2014
Time Allotted: 2:30 hrs

Name: _____ ID Number _____

INSTRUCTION 1: READ THE FOLLOWING QUESTIONS AND CHOOSE THE BEST ANSWER AND WRITE THE LETTER OF YOUR CHOICE ON THE ANSWER SHEET PROVIDED, ATTACHED AT THE LAST PAGE.

PART I: GROSS ANATOMY (16 QUESTIONS)

1. Which of the following vertebral body is heart shaped and contains one or two costal facets?
 - a. Lumbar
 - b. Thoracic
 - c. Cervical
 - d. Sacral
 - e. All of the above
2. Which of the following is **Wrong** about the intervertebral foramen?
 - a. Formed by the vertebral notch of adjacent vertebral pedicles
 - b. An exit for spinal nerves from the spinal canal
 - c. Its posterior border is formed by the zygapophysial joints of the articular processes.
 - d. A herniated intervertebral disc can result narrowing of the intervertebral foramen and may compress the spinal nerve roots.
 - e. None of the above
3. Which of the following is **not** from characteristic features of the extrinsic back muscles?
 - a. They are innervated by the posterior rami of segmental spinal nerves
 - b. They are innervated by the anterior rami of segmental spinal nerves
 - c. The intermediate group muscles have respiratory function
 - d. The superficial group muscles move the upper limb
 - e. None of the above
4. Orientation of the zygapophysial joints of the back show differences that results differences in the degrees of movement in the different regions of the back. Which of the following describes the orientation and movement in the lumbar region?
 - a. Vertical orientation that limits flexion and extension, but facilitate rotation
 - b. Interlock, thereby limiting range of movement
 - c. slope inferiorly from anterior to posterior and facilitate flexion & extension
 - d. Vertical orientation with limiting range of movement
 - e. Horizontal orientation that facilitates rotation

5. Which of the following is **not** muscle of the axial skeleton?
- a) Zygomaticus major muscle
 - b) Brachialis muscle
 - c) Levator ani muscle
 - d) Quadratus lumborum muscle
 - e) None of the above
6. Which of the following is a **wrong** matching between the injury and the affected muscle?
- a. Thoracodorsal nerve injury----- weakness in latissimus dorsi muscle
 - b. Dorsal scapular nerve injury----- weakness in rhomboid minor muscle
 - c. Accessory spinal nerve injury ---- Weakness in rhomboid major muscle
 - d. Injury to cranial nerve eleven (CN XI) ----- Weakness in trapezius muscle
 - e. Dorsal scapular nerve injury ----- Weakness in rhomboids muscles
7. **Wrong** statement about intervertebral disc (IV)?
- a. Posterolateral herniations of the nucleus pulposus are most common in the lumbar and
 - b. Degenerated annulus fibrosus allows herniation of the nucleus pulposus into the vertebral canal
 - c. The outer cartilaginous ring of the intervertebral disc is formed by annulus fibrosus
 - d. A herniated IV disc that compresses the spinal nerve roots results localized pain.
 - e. None of the above
8. All the following muscles are found in the lower limb **except one**
- a) Biceps femoris muscle
 - b) Tibialis posterior muscle
 - c) Sartorius muscle
 - d) Abductor pollicis longus muscle
 - e) None of the above
9. All the following muscles are found in the upper limb **except one**?
- a) Supinator muscle
 - b) Dorsal interosseous muscle
 - c) Extensor digitorum brevis muscle
 - d) Pronator teres muscle
 - e) Palmar interosseous muscle

10. All the following arteries are found in the abdominal cavity except one?

- a) Superior phrenic arteries
- b) Middle suprarenal arteries
- c) Testicular arteries
- d) Ovarian arteries
- e) None of the above

11. Venous drainage from the head and neck could be by all the following veins except one?

- a) Vertebral veins
- b) Internal jugular veins
- c) Dural sinuses
- d) Subclavian veins
- e) None of the above

12. Which type of cartilage is found in the walls of the eustachian tube?

- A. Elastic cartilage
- B. Hyaline cartilage
- C. Fibrocartilage
- D. All of the above
- E. None of the above

13. Multinucleated cells involved in the resorptions and remodeling of bone tissue

- A. Osteocytes
- B. Osteoblasts
- C. Osteoclasts
- D. Osteoprogenitor cells
- E. Osteon

14. Which type of cartilage forms the epiphyseal growth plate?

- A. Hyaline cartilage
- B. Elastic cartilage
- C. Fibrocartilage
- D. All of the above
- E. None of the above

15. Cell bodies of motor neurons associated with spinal motor nerve roots may be located:

- a. Within the epimysium of the muscle they innervate.
- b. In anterior gray horn of the spinal cord
- c. In dorsal root ganglia.
- d. At any point along a motor nerve.
- e. In the cerebral cortex.

16. In which level of the vertebrae the spinal cord is located by the 6th month of the intrauterine life
- Extends the entire length of embryo
 - At the level of S.1
 - At the level of L2 or L3
 - At the level of lower border of L1
 - At the level of T12

PART II: GENERAL HISTOLOGY (30 QUESTIONS)

17. Following Haematoxylin & Eosin (H & E) staining, the eosinophilic components of the tissue:
- Stain red-pink
 - Include the nucleus
 - May show metachromatic reaction
 - Stain the ribosomes
 - Are acidic components of the tissue
18. The final magnification of tissues you see under the light microscope may be found:
- By the product of magnifying powers of condenser lens and objective lens.
 - By the product of magnifying powers of condenser lens and eye piece lens.
 - By the product of magnifying powers of objective lens and eye piece lens.
 - By the product of magnifying powers of condenser lens, objective lens & eye piece lens.
 - Only as the magnifying power of the objective lens.
19. The plasma membrane:
- Has peripheral proteins which are mostly found loosely attached on the outer surface of the membrane.
 - Has integral proteins which show free lateral mobility and rearrangements.
 - Has oligosaccharides which are found projecting from both outer and inner surfaces.
 - Is involved in signal transduction through different classes of polysaccharides receptors.
 - Is clearly visible under light microscopy.
20. Centrioles are constructed from:
- Lysosomes
 - Peroxisomes
 - Microfilaments
 - Microtubules
 - Intermediate filaments
21. Organelles involved in the synthesis of polysaccharides, and glycosylate proteins and lipid and make them ready to be secreted are:
- Rough endoplasmic reticulum
 - Smooth endoplasmic reticulum
 - Peroxisomes
 - Golgi bodies
 - Lysosomes
22. Which of the following organelles are used for intracellular transport system?
- Microtubules
 - Peroxisomes
 - Microfilaments
 - Smooth endoplasmic reticulum.
 - Intermediate filaments

16. In which level of the vertebrae the spinal cord is located by the 6th month of the intrauterine life
- Extends the entire length of embryo
 - At the level of S.1
 - At the level of L2 or L3
 - At the level of lower border of L.1
 - At the level of T12

PART II: GENERAL HISTOLOGY (30 QUESTIONS)

17. Following Haematoxylin & Eosin (H & E) staining, the eosinophilic components of the tissue:
- Stain red-pink
 - Include the nucleus
 - May show metachromatic reaction
 - Stain the ribosomes
 - Are acidic components of the tissue
18. The final magnification of tissues you see under the light microscope may be found:
- By the product of magnifying powers of condenser lens and objective lens.
 - By the product of magnifying powers of condenser lens and eye piece lens.
 - By the product of magnifying powers of objective lens and eye piece lens.
 - By the product of magnifying powers of condenser lens, objective lens & eye piece lens.
 - Only as the magnifying power of the objective lens.
19. The plasma membrane:
- Has peripheral proteins which are mostly found loosely attached on the outer surface of the membrane.
 - Has integral proteins which show free lateral mobility and rearrangements.
 - Has oligosaccharides which are found projecting from both outer and inner surfaces.
 - Is involved in signal transduction through different classes of polysaccharides receptors.
 - Is clearly visible under light microscopy.
20. Centrioles are constructed from:
- Lysosomes
 - Peroxisomes
 - Microfilaments
 - Microtubules
 - Intermediate filaments
21. Organelles involved in the synthesis of polysaccharides, and glycosylate proteins and lipid and make them ready to be secreted are:
- Rough endoplasmic reticulum
 - Smooth endoplasmic reticulum
 - Peroxisomes
 - Golgi bodies
 - Lysosomes
22. Which of the following organelles are used for intracellular transport system?
- Microtubules
 - Peroxisomes
 - Microfilaments
 - Smooth endoplasmic reticulum
 - Intermediate filaments

23. The outer membrane of the nucleus envelop is continuous with
- Microfilaments
 - Peroxisomes
 - Rough endoplasmic reticulum
 - Goli bodies
 - Smooth endoplasmic reticulum
24. Regarding to the nucleolus
- Is synonymous of the nucleus
 - Is the site of mitochondrial synthesis
 - Is membrane bound region within the nucleus
 - Is made by regions of chromatin contributed by different chromosomes
 - Is not visible under the light microscope
25. During a cell cycle,
- Duplication of centrioles begins and complete within the G_1 phase
 - The S phase is the longest and most variable phase
 - RNA and protein synthesis occur during the G_1 phase
 - Synthesis of DNA occurs during the S phase
 - Replication of the chromosomes occurs during G_1 phase
26. An intermediate filament present in the nuclear lamina and used for organization of the nuclear membrane and perinuclear chromatin is
- Vimentin
 - Desmin
 - Glial fibrillary acid protein
 - Cytokeratin
 - Lamins
27. Which of the following is not inclusion?
- Lipid droplets
 - Endosomes
 - Glycogen granules
 - Lipofuscin
 - Melanin
28. An electron microscope is needed to observe all the following features of epithelia except one?
- Zonula occludens
 - Hemidesmosomes
 - Basement membrane
 - Cilia
 - Flagella
29. Which of the following is a wrong statement about gap junctions?
- Each gap junction could be made up of more than two clusters of connexons
 - Each connexon is made up of only six connexins
 - Each connexon/cating channel is made up of pairs of connexins
 - Each connexon protein in a given gap junction could have a different chemical property than its neighbor
 - None of the above

23. The outer membrane of the nuclear envelop is continuous with:
- Microfilaments
 - Peroxisome
 - Rough endoplasmic reticulum
 - Golgi bodies
 - Smooth endoplasmic reticulum
24. Regarding to the nucleolus:
- It is synonymous of the nucleus.
 - Is the site of mitochondrial synthesis
 - Is membrane bound region within the nucleus.
 - Is made by regions of chromatin contributed by different chromosomes
 - Is not visible under the light microscope
25. During a cell cycle,
- Duplication of centrioles begins and complete within the G_2 phase.
 - The S phase is the longest and most variable phase.
 - RNA and protein synthesis occur during the G_2 phase
 - Synthesis of DNA occurs during the S phase.
 - Replication of the chromosomes occurs during G_2 phase.
26. An intermediate filament present in the nuclear lamina and used for organization of the nuclear membrane and perinuclear chromatin is:
- Vimentin
 - Desmin
 - Glia fibrillary acid protein
 - Cytokeratin
 - Lamins
27. Which of the following is not inclusion?
- Lipid droplets
 - Endosomes
 - Glycogen granules
 - Lipofuscin
 - Melanin
28. An electron microscope is needed to observe all the following features of epithelia except one?
- Zonulaoccludens
 - Hemidesmosomes
 - Basement membrane
 - Cilia
 - Flagella
29. Which of the following is a wrong statement about gap junctions?
- Each gap junction could be made up of more than ten numbers of connexions
 - Each connexon is made up of only six connexons
 - Each connexon-cating channel is made up of pairs of connexons
 - Each connexon protein in a given gap junction would have a different chemical property than its neighbor
 - None of the above

- c) In ordinary connective tissue, the predominant component of ground substance is water.
- d) Fibroblast produces and maintains all extracellular matrix components of connective tissue.
- e) None of the above.
37. Concerning the **connective tissue**, which statement is *false*?
- a) Proteoglycans are responsible for the highly viscous character of the ground substance.
- b) In blood, the ground substance lacks stabilizing macromolecules.
- c) Reticular fibers are particularly abundant in the framework of hematopoietic organs.
- d) Macrophages are also antigen-presenting cells.
- e) None of the above.
38. Which *connective tissue cell type* is functionally related to the *basophils*?
- a) Fibroblast
- b) Mast cell
- c) Pericyte
- d) Plasma cell
- e) Myofibroblast
39. With regard to **cartilage**, which statement is *false*?
- a) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- b) Hyaline cartilage is characterized by an amorphous glassy matrix.
- c) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- d) Fibrocartilage is used for endochondral bone ossification.
- e) Fibrocartilage is present in regions subjected to high tension.
40. **Elastic cartilage**.
- a) is found in the annulus fibrosus and intervertebral disks.
- b) the collagen in its matrix is mainly type II.
- c) is the primary skeletal tissue in the fetus.
- d) has no identifiable perichondrium.
- e) is the most widely distributed cartilage type in the body.
41. With regard to **bone**, which statement is *true*?
- a) Bone formation without a pre-existing cartilage model is called endochondral bone formation.
- b) Interstitial lamellae represent smooth layers of bone on the surfaces of joints.
- c) Compact (dense) bone is characterized by the presence of osteons.
- d) Chondroblasts secrete organic components of bone matrix.
- e) Osteoblasts are involved in bone resorption.
42. Which of the following statements is *correct with* regard to the **bone morphology**?
- a) Primary bone has a higher mineral content than the secondary bone.
- b) Haversian's lacunae are calcified collagen bundles of the periosteum that run in concentric manner.
- c) Osteoclasts have processes that run in the form of canaliculi.
- d) Osteoblasts are derived from blood monocytes.
- e) None of the above.

43. Which of the following cells *does not* belong to the *mononucleated phagocytic system*?
- a) Microglia
 b) Hepatocytes
 c) Dust cells
 d) Langerhan's cells
 e) Kupffer cells

44. With regard to the *white blood cells*, which of the following statements is *true*?
- a) Neutrophils are the most common white blood cells in circulating blood.
 b) A patient with a parasitic infection is likely to reveal an increase in the circulating monocytes.
 c) Basophils normally re-enter the bloodstream after leaving circulation and immigrating into peripheral tissues.
 d) A macrophage can originate from a mast cell.
 e) None of the above.

45. Fibrocartilage is not found in the:

- a) Intervertebral discs
 b) Pubic symphysis
 c) Discs of knee joint
 d) Temporomandibular joint
 e) Epiglottis

46. Select the *incorrect statement* about *mature erythrocytes*.

- a) do not contain nuclei.
 b) possess haemoglobin, which fills almost the entire cytoplasm.
 c) are unable to move actively, but they are remarkably elastic and can withstand deformation.
 d) contain the organelle peroxisome as oxygen binding protein.
 e) None of the above.

PART III: Biochemistry (17 questions)

47. Lateral diffusion of membrane lipids occurs at a very slow rate.

- a) True
 b) False

48. Phosphatidic acid is a common structural component of different phospholipids.

- a) True
 b) False

49. A membrane rich in saturated fatty acids is more fluid than the one rich in unsaturated fatty acids.

- a) True
 b) False

50. Sucrose and lactose are examples of reducing disaccharides.

- a) True
 b) False

51. Hyaluronic acid serves as lubricant and shock-absorber in synovial fluids.

- a) True
 b) False

52. Naturally occurring fatty acids usually contain *trans* double bonds.

- a) True
 b) False

53. Fats and oils are acylglycerols that differ in the nature of their fatty acids.

- a) True
 b) False

54. N-acetylglucosamine is a common component of ABO blood group determinants.

- a) True
 b) False

55. Which type of amino acid serves as a precursor for the biosynthesis of niacin (vitamin B3) in our body?
- A. Glycine
 - C. Tryptophan
 - D. Cysteine
 - E. Lysine
56. Cysteine and methionine
- A. Contain sulfur
 - B. Have polar side chains
 - C. Have non-polar side chains
 - D. Contain iminio group
 - E. A & C
57. A tripeptide that protects cell membranes from damage with its antioxidant properties is
- A. Corticotropin
 - B. Vasopressin
 - C. Bradykinin
 - D. Angiotensin
 - E. Glutathione
58. Collagen contains hydroxylated derivatives of
- A. Arginine and glutamate
 - B. Cysteine and tryptophan
 - C. Lysine and proline
 - D. Tyrosine and dopamine
 - E. Valine and isoleucine
59. One of the following proteins differs in architecture and solubility from the others
- A. Immunoglobulin
 - B. Keratin
 - C. Elastin
 - D. Myosin
 - E. Fibrinogen
60. Which one of the following could not be found as a component of nucleic acids?
- A. Uracil
 - B. Hypoxanthine
 - C. Uric acid
 - D. Xanthine
 - E. B and D
61. The important functional groups in purines and pyrimidines for the formation of base pairs include
- A. The amino groups of guanine and cytosine
 - B. The oxygen linked to the fourth carbon of uracil and thymine
 - C. The third nitrogen of pyrimidine bases
 - D. The first nitrogen of purine bases
 - E. All of the above

62. Which nucleotide serves as a second messenger in signal transduction pathways?

- A. Cytidine diphosphate
- B. Cyclic guanosine monophosphate
- C. Cyclic adenosine diphosphate
- D. Cyclic uridine diphosphate
- E. Cyclic adenosine monophosphate

63. Which one of the following statements regarding the Watson-Crick DNA model is incorrect?

- A. Polynucleotide chains are anti-parallel with each other
- B. Sugar-phosphate chains occupy the core of the double helix
- C. Polynucleotide chains wind around a central axis
- D. Base pairs are stabilized through hydrogen bonding
- E. The double strands of DNA form a right-hand

PART IV: GENERAL EMBRYOLOGY (19 QUESTIONS)

64. Which of the following is the result of spermatogenesis at meiosis II?

- a) Produces two secondary spermatocytes, haploid single structured chromosomes
- b) Produces two secondary spermatocytes, diploid double structured chromosomes
- c) Produces four spermatids, haploid double structured chromosomes
- d) Produces four spermatids, haploid single structured chromosomes
- e) None of the above

65. Which of the following is not part of spermiogenesis?

- a) Condensation of the nucleus and acrosome formation
- b) Shedding of most of the cytoplasm
- c) Entering a prolonged prophase of 22 days
- d) Formation of neck, middle piece and tail
- e) None of the above

66. The _____ is a derivation of cumulus oophorus formed following ovulation of the oocyte.

- a) Corona radiata
- b) Zona pellucid
- c) Theca interna
- d) Polar body
- e) Theca externa

67. Which of the following is not related with a surge of luteinizing hormone during the midovarian cycle?

- a) Causes follicular rupture and ovulation
- b) Stimulates production of progesterone by follicular stromal cells
- c) Causes thinning of cervical mucus
- d) Elevates concentrations of maturation promoting factor
- e) None of the above

68. Which of the following structure among the given alternative comes last developmentally

- A. Epiblast
- B. Secondary yolk sac
- C. Chorionic cavity
- D. Amnion
- E. Prechordal plate

69. Extraembryonic mesoderm is derived from cell arising from

- A. Amnion
- B. Heuser's membrane
- C. Epiblast
- D. Hypoblast
- E. Cytotrophoblast

70. Which of the following is not the function of notochord
- A. Defines the primordial longitudinal axis of the embryo
 - B. Provides signals that are necessary for the development of axial musculoskeletal
 - C. Degenerates as the bodies of the vertebrae form, but small portions of it persist as the *Anulus fibroxus* of each intervertebral disc.
 - D. Induces the overlying embryonic ectoderm to thicken and form the neural plate
 - E. None
71. Which of the following tissue is derived from the surface ectoderm
- A. Anterior lobe of the pituitary gland
 - B. Retina of the eye
 - C. The pia and arachnoid mater
 - D. Multipolar motor neuron
 - E. The neurons of the sensory ganglia of the cranial nerves
72. The main event which takes place by the seventh week of the developing individual is/are
- A. Scalp vascular plexus
 - B. Ossification of bone in the upper limb has begin
 - C. Appearance of auricular hillocks
 - D. Formation of Mesonephric ridges
 - E. Appearance of the fourth pharyngeal arch
73. Which of the following is appropriate diagnostic techniques for accessing the fetal chromosomal abnormalities, inborn errors of metabolism, and X-linked disorders
- A. Spectrophotometric Studies
 - B. Chorionic Villus Sampling
 - C. Sex Chromatin Patterns
 - D. Cell Cultures and Chromosomal Analysis
 - E. Percutaneous Umbilical Cord Blood Sampling
74. All are events occurring during the pre embryonic period of the prenatal development except:
- A. Formation of neural tube.
 - B. Formation of heart tube from the cardiogenic mesoder.
 - C. Formation of stomodinium and proctodinium.
 - D. Formation tertiary chorionic villa.
 - E. None of the above
75. Concerning the **fetal membranes**, which statement is **true**?
- a) The yolk sac helps control the embryo's body temperature.
 - b) As the bladder enlarges, the umbilical cord involutes to form urachus.
 - c) Blood formation occurs in the wall of allantois during the third to fifth weeks.
 - d) Primordial germ cells appear in the endodermal lining of the wall of the allantois.
 - e) During the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.

76. Concerning **multiple pregnancies**, which statement is **false**?
- The frequency of monozygotic twinning shows marked racial differences.
 - The rate of dizygotic twinning increases with maternal age.
 - Dizygotic twins always have two amnions and two chorions.
 - Dizygotic twinning shows a hereditary tendency.
 - None of the above.
77. **Diethylstilbestrol (DES)** was once prescribed to women to prevent miscarriages. However, it has since been found to be a teratogen that causes what type of defects?
- Reproductive system
 - Central nervous system
 - Gastrointestinal
 - Limb (partial or total absence)
 - Heart and neural tube defects
78. Most **birth defects** occur during:
- The first 2 weeks
 - The embryonic period
 - The fetal period
 - Early childhood
 - None of the above
79. With regard to **birth defects**, which statement is **false**?
- Clubfoot, which is due to mechanical forces in the womb, is an example of deformation.
 - Trisomy of the autosomes occurs with increasing frequency as maternal age increases.
 - Deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
 - The phenotype of Turner syndrome is female.
 - Thalidomide induces limb defects.
80. Which of the following is **not the function of the amniotic fluid**?
- Acts as a barrier to infection
 - Permits normal fetal lung development
 - Prevents adherence of the amnion to the embryo and fetus
 - Cushions the embryo and fetus against injuries by distributing impacts the mother receives
 - None of the above.
81. The pericardioperitoneal canals develop into:
- Pericardial cavity
 - Peritoneal cavity
 - Pleural cavities
 - a & b
 - All of the above
82. With regard to the development of diaphragm:
- The central tendon of the diaphragm develops from dorsal mesentery of the esophagus.
 - Septum transversum gives rise to the couple of the diaphragm.
 - The costal margin of the diaphragm develops for the pleuropericardial folds.
 - Diaphragm ascends during development.
 - Eventration of the diaphragm could lead to hypoplasia of the lung.

PART V: PHYSIOLOGY (21 QUESTIONS)

83. Symport may be:
- Na^+ -glucose transport
 - Cl^- - HCO_3^- transport
 - Ca^{++} - Na^+ transport
 - Na^+ - K^+ transport
 - None of the above

84. The cell is characterized by
- Self-preservation
 - Reproduction
 - A 10-20% protein mass
 - The presence of sex-chromatin in its nucleus
 - All of the above
85. Which of the following is the fattest individual?
- An infant with body water content of 65%
 - An infant with body water content of 80%
 - An adult with body water content of 45%
 - An adult with body water content of 60%
 - None of the above
86. A drop in interstitial fluid (IF) volume leads to:
- Low BP
 - Poor skin turgor
 - Dry tongue
 - Rapid pulse
 - (b) and (c) above
87. Which of the following is indicative of heavy exercise?
- A urine output of 500 ml/day
 - A urine output of 1200ml/day
 - A sweat output of 1400ml/day
 - A sweat output of 5L/ day
 - "a" and "d" above
88. Causes of pitting oedema include all of the following except:
- Increased hydrostatic pressure in capillaries secondary to heart failure
 - Decreased osmotic pressure due to low plasma proteins secondary to conditions like nephrotic syndrome
 - Increased capillary permeability.
 - Depression of cellular metabolism
 - Localized Increased venous pressure
89. All of the following **except one** are examples of positive feedback mechanism
- Progress of labour
 - Nerve impulse
 - Release of cortisol
 - Replication of viruses
 - Irreversible shock
90. One of the following serves as a detector in the regulation of BP:
- Vasomotor centre
 - Arteriole
 - Aortic body
 - Carotid artery
 - Aortic arch